Robot Status

Recorded condition and recommended next steps

Review details	
Client	Example client · Synthetic demonstration
Robot / asset	Packaging cell 2
Prepared by	Service review
Robot serial	SYNTHETIC-DEMO
Robot model	Not established
Controller software	5.24
Incident dates	2026-10-12
Report generated	2026-11-02 15:39 UTC

Attention required

Controller evidence identifies concerns involving Base, Elbow. These are the first areas to investigate; the evidence does not establish that the joints need replacement.

Start with the recorded process stage, physical clearance and active load settings. Use the joint summaries to distinguish application-related warnings from controller messages that require service interpretation.

Evidence is insufficient to characterize Shoulder, Wrist 1, Wrist 2, Wrist 3. These joints remain unassessed or only partly observed; absence of a recorded fault is not a clean bill of health.

Source coverage has qualifications. Resolve missing or omitted evidence before drawing conclusions about fault-free operation.

This assessment describes the available records. It is not an inspection of the robot’s present mechanical condition or a safety approval.

Local evidence-based review; hypotheses require verification. Controller detections do not alone prove physical contact or component failure. Not a safety validation.

Joint condition

Each joint is assessed separately. “Not assessed” and “Limited evidence” are gaps in the available records, not a pass or a prediction of remaining life.

Assessment by joint	
Base	Needs attention The controller recorded collision detection or path-following warnings for Base. The associated recording also contains current outside recorded bounds; review the load and motion context. These findings do not establish that the joint is damaged.
Shoulder	Limited evidence No mapped warning naming Shoulder was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Elbow	Needs attention The controller recorded abrupt torque-demand warnings for Elbow. No matched joint telemetry was available to corroborate the warnings. These findings do not establish that the joint is damaged.
Wrist 1	Limited evidence No mapped warning naming Wrist 1 was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Wrist 2	Limited evidence No mapped warning naming Wrist 2 was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Wrist 3	Limited evidence No mapped warning naming Wrist 3 was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.

Recommendations below connect the observed concerns to practical checks. Full incident measurements can be included in the optional technical evidence appendix.

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Program areas to review

Approach path and clearance

Program context	
Program	Synthetic_Pickup
Review priority	Investigate
Evidence association	Matched incident recording; marker locates recorded event, not proven root cause

The controller reported collision detection or a path-following discrepancy. The approach, retract path and external cabling are relevant review areas, together with the active load model.

Recorded script context: lines 42–43. These are execution references, not proof that the line caused the issue.

Supporting cases: INC-001

Move transitions and blending

Program context	
Program	Program not identified
Review priority	Investigate
Evidence association	No matching captured program context

The controller reported abrupt or excessive torque demand. Review the transition between moves and commanded acceleration before treating this as a hardware problem.

Supporting cases: INC-002

Recommended action plan

Recommended next steps

1. Review Base, Elbow at the affected motion stage: check physical clearance, active payload and centre of gravity, then acceleration changes and move transitions.

2. Inspect the real cell at the recorded approach/retract stage and compare fixture, workpiece and cable clearance with the intended path.

3. Inspect acceleration changes, blend settings and required stops around the recorded motion. Validate any change against clearance and process requirements.

4. Obtain the matching flight recording, captured script and installation before assigning this concern to a specific program area or line.

5. Collect representative controller logs and joint motion recordings for the insufficiently covered joints, and confirm current condition with the maintenance history and physical inspection.

Verify the application and physical setup before changing motion or load settings. Retain the original evidence for follow-up and service review.

Scope of this assessment

Reviewed 1 controller log dataset(s) and 1 flight recording(s).

The findings describe the supplied history and recorded windows, not continuous monitoring or a physical inspection.

Available incident dates: 2026-10-12.

Source warnings were reported. Check the technical coverage details before treating this as a complete review.

Coverage notes

Synthetic demonstration data; not a real robot diagnosis.

Evidence still needed

This review does not predict remaining joint life or certify that the robot is safe to operate. Findings must be considered with the real application and any maintenance completed after these records.

Recommended action plan (continued)

Technical references

1. UR technical guidance
https://www.universal-robots.com/articles/ur/robot-care-maintenance/understanding-protective-stops/

2. UR technical guidance
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-err-codes/topics/CODE_153.html

3. UR technical guidance
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-err-codes/topics/CODE_157.html

4. UR technical guidance
https://www.universal-robots.com/manuals/EN/HTML/SW5_19/Content/prod-err-codes/topics/CODE_174.html

5. UR technical guidance
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-err-codes/topics/CODE_174.html

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